Network Troubleshooting: Socket Listen State & Interface Bind Address

Duration : 30 Min.

Scenario

A mission-critical payment processing host is reported 'Down' by synthetic monitors. The application support team checks the server and reports: 'The payment process Java PID 4821 is running and using CPU, so the service is fine.' Yet remote clients cannot connect on port 8080. The network engineer captured `ss -tulpn` output and an external client connection log. The engineer must inspect these artifacts, identify the bind address anomaly, and formulate the remediation.

Provided Artifacts & Input Files

The following diagnostic artifacts are pre-provisioned inside your `student_workspace/` directory for your inspection and analysis:

- `application_config.properties`: Service deployment and network interface binding configuration.
- `external_connection_attempt.log`: Client-side connection attempt trace recording connection status.
- `ss_tulpn_capture.txt`: Linux socket statistics (`ss -tulpn`) capturing listening ports and bind addresses.
- `triage_report_template.md`: Starter template to record your technical analysis, findings, and remediation plan.

Task Objectives

Perform the following network troubleshooting diagnosis and document your findings:

1. Open and inspect `ss_tulpn_capture.txt`, `external_connection_attempt.log`, and `application_config.properties` in `student_workspace/`.
2. Analyze the socket listening state, protocol, port, and bind address of the running service process.
3. Explain the network stack behavior and kernel response when external traffic attempts to reach a service bound to a specific local address.
4. Identify the root cause of the connection refusal and determine the required configuration update.
5. Document your technical analysis and configuration change in `student_workspace/triage_report.md`.

Instructions to Perform the Task

1. Review the scenario and inspect the pre-provisioned artifact files in your `student_workspace/`.
2. Formulate your technical diagnosis, packet/route analysis, and remediation plan based on the artifact evidence.
3. Save your comprehensive analysis inside `student_workspace/triage_report.md` (using `triage_report_template.md` as your guide).
4. Review your submission against the **Evaluation Rubric** below to ensure all technical criteria are addressed.
5. When complete, return to the platform dashboard and click **"Submit Final"**. This assessment is evaluated manually by the evaluation team.

Validation

Once your deliverable is prepared and saved in the workspace, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

Item	Requirement / Criteria	Marks
Rubric 1	Accurate interpretation of socket tables, listening states, and interface bind semantics	4 Marks
Rubric 2	Technical explanation of kernel TCP response behavior for unexposed interfaces	3 Marks
Rubric 3	Formulation of the exact configuration update required to enable external connectivity	3 Marks

Total Score: 10 Marks

Important Notes

- This is a **Manual Evaluation** assessment.
- Focus on sound networking principles, accurate protocol diagnosis, and structured remediation.